

Task 16 — Hypothesis Testing & A/B Tests

Data Analyst · Phase 1 Industry Immersion · Study Guide

DATA ANALYST

PlaceMux · Phase 1 · Study Guide

DIFFICULTY

Advanced · Advanced

Level 16 of 20 · ramp 76%

1 · What this is & why it matters

Design and run hypothesis tests / A-B experiments to decide whether an observed difference is real or chance.

Decisions based on noise waste money. Proper testing separates real effects from random fluctuation.

2 · Learning objectives

By the end of this guide and task you will be able to:

- Deliver: A documented test/experiment with effect size, significance and a recommendation.
- Explain the key concepts behind the task in your own words.
- Choose between alternative ways to execute it, and justify your choice.
- Avoid the common pitfalls and know how your work is scored.

3 · Prerequisites

- A reproducible Python data environment (pandas, numpy, matplotlib).
- Comfort reading data into a DataFrame and writing basic SQL.
- The habit of validating data before analysing it.

4 · Key concepts (understand these first)

Null vs alternative hypothesis

the default 'no effect' vs the claim.

p-value & significance

the chance of seeing this if nothing's really going on.

Effect size

how big the difference is, not just whether it's significant.

Sample size & power

enough data to detect a real effect.

Multiple-testing risk

test enough things and something looks 'significant' by luck.

5 · Recommended stack & tools

- scipy.stats / statsmodels
- pandas
- a power calculator

6 · The build pipeline (follow in order)

This is the flow from nothing to demoable. Each step builds on the last; the final steps verify and prove the work.

1. State the hypothesis and the decision it informs.

2. Choose the right test and significance level up front.
3. Estimate required sample size/power.
4. Run the test on clean, comparable groups.
5. Report effect size with the p-value and confidence interval.
6. Translate the result into a clear recommendation.

7 · Other ways to execute (alternative approaches)

There is rarely one right path. Consider these alternatives and pick deliberately for your context:

- Frequentist A/B vs Bayesian A/B.
- t-test vs non-parametric tests for skewed data.

8 · Brainstorming & deeper thinking

Sit with these questions — they separate someone who completed the task from someone who understood it:

- What confound could explain the difference instead of your variable?
- Is the effect size big enough to actually act on?
- Did you decide the test before peeking at the data?

9 · Definition of Done

Done when all are true and demoable

- A documented test/experiment with effect size, significance and a recommendation.
- Demonstrable live on real data.

10 · How your work is evaluated (out of 100)

Scoring parameter	Marks
Core deliverable — A documented test/experiment with effect size, significance and a recommendation.	50
Real-data quality & correctness (real inputs at realistic scale, not a toy/happy-path)	20
Live verification & evidence (demonstrated live; real output, not claims)	15
Dependency, failure & edge-case handling (errors handled; hand-offs honoured)	15
TOTAL	100

11 · Pitfalls to avoid

Worry if any of these are true of your work

- p-hacking by testing until something sticks.
- Significant but trivially small effects.
- Unequal or contaminated groups.

12 · Where this fits & go deeper

Over Phase 1 you grow from loading a CSV to shipping a live, decision-grade analytics product. You turn raw data into numbers people trust and decisions people make.

Go deeper (optional further study):

- Statistics for analysts (distributions, tests, intervals)
- SQL window functions and query optimisation

- Storytelling with data and dashboard design
- Time-series and segmentation techniques